

Mind map of our algebra course.

Field k ($\mathbb{Q}, \mathbb{R}, \mathbb{C}$, finite field)

$\mapsto k[x_1, \dots, x_n]$ polynomial ring

Particular case $n=1$: $k[x]$ more special
kind of similar to $\mathbb{Z}$

$f_1, \dots, f_m \in k[x_1, \dots, x_n] \mapsto$ understand the solution of the system

$$\begin{cases} f_1 = 0 \\ \vdots \\ f_m = 0 \end{cases}$$

solve over k

$V_k(f_1, \dots, f_m)$
variety

is it finite?
How can I get to the solution?
What is the dimension?
What is the degree?

$I = \langle f_1, \dots, f_m \rangle$ ideal

$V_k(I)$ ideal gives the variety

We had algorithms and structural results

- Hilbert basis theorem
- Gröbner bases
- Elimination of variables

Quotient rings
 R/I
Chinese Remainder Theorem, very general version.

Algebra
ideals in a unitary commutative ring (this is what we did)

Geometry
Varieties.

Algebraic geometry

There is quite a difference between a field k and its algebraic closure $\bar{k}$.

For example, you want to understand $V_k(I)$ as this might be tricky, whereas $V_{\bar{k}}(I)$ is more general.

Shiquin: Gordan-Dickson lemma.

Marharyta: Hilbert basis theorem

Rushi: Ideals in $K[x]$. (14.1, 1.5)

Hajji: CRT $\Rightarrow$ RSA

Dickson's Lemma!

for $\forall A \subseteq \mathbb{Z}_{\geq 0}^n$, $\exists B \subseteq A$, ^{finite} s.t. $\langle x^\alpha : \alpha \in A \rangle = \langle x^\beta : \beta \in B \rangle$

$x \in A$
 x element of A

$A \subseteq B$

A is a subset of B

Example

$1 \in \mathbb{R}$

$1, 2, 3 \in \mathbb{R}$

Lemma (Equality test for monomial ideals)

Let $A, B \subseteq \mathbb{Z}_{\geq 0}^n$. Then

$$\langle x^\alpha : \alpha \in A \rangle = \langle x^\beta : \beta \in B \rangle \iff A + \mathbb{Z}_{\geq 0}^n = B + \mathbb{Z}_{\geq 0}^n$$

Lemma (Description of monomial ideals).

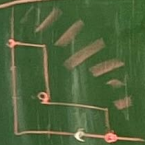
for $A \subseteq \mathbb{Z}_{\geq 0}^n$, $\langle x^\alpha : \alpha \in A \rangle = \text{lin}_k \{ x^\beta : \beta \in A + \mathbb{Z}_{\geq 0}^n \} = \left\{ \sum_{\beta \in A + \mathbb{Z}_{\geq 0}^n} c_\beta x^\beta : c_\beta \in k \text{ for any } \beta \in A + \mathbb{Z}_{\geq 0}^n \text{ and } c_\beta = 0 \text{ for any other } \beta \right\}$

$\text{lin}_k = \text{span}_k$

Example

$$\langle x^3, xy, y^3 \rangle = \left\{ \frac{f(x,y)}{xy} : f(x,y) \in k[x,y]^3 \right\}$$

$$\langle x^3, xy \rangle$$



$$\begin{aligned} b_1 &= x^2y^2 - 3xy^2 + y^5 + x^3y^3 \\ b_2 &= 3xy^3 - y^5 + x^3y^3 \end{aligned}$$

$$\begin{cases} f_1 = x^2y^2 \\ f_2 = x + xy^2 \end{cases}$$

$$f = (b_1, b_2)$$

$$\begin{aligned} ① \quad a+b &= b+a \\ ② \quad (a+b)+c &= a+(b+c) \\ ③ \quad a \cdot (b+c) &= a \cdot b + a \cdot c = b+a \cdot c \end{aligned}$$

$$④ \quad a \cdot b = b \cdot a$$

$$⑤ \quad (a \cdot b) \cdot c = a \cdot (b \cdot c)$$

$$⑥ \quad 0 + a = a$$

$$⑦ \quad a \cdot 1 = a$$

$$⑧ \quad a \in (\mathbb{R}, +) \Leftrightarrow -a \in (\mathbb{R}, +)$$

$$a \text{ or } ? = a$$

$$\forall a \in \mathbb{R} \exists b. a+b=0$$

there is to
be unique for
a given a

$\mathbb{Z}$ example of a unitary
commutative (b)
ring.

$\mathbb{R}^{2 \times 2}$ a ring,
unitary,
not commutative

$\begin{aligned} + &= \text{or} \\ \cdot &= \text{and} \end{aligned} \left. \begin{aligned} &\text{both conditions} \\ &\text{are satisfied} \end{aligned} \right\}$
 $R = \{f, b, true\}$

wrong example

$(\mathbb{Z} \geq 0, +, \cdot)$ satisfies ①-⑦
but still not a ring.

false or b =

very bad $0 \neq 1$

Type some Sage code below and press Evaluate.

```
1 R = PolynomialRing(QQ, 'x,y')
2 x,y = R.gens()
3 I = R.ideal(x^3, x*y, y^3)
4 f = - 3*x*y^2 + y^5 + x^3 * y^3
5 g = x^2 + y^2
6 h = x*y^2
7 f in I
```

Evaluate

True